

Google Associate Cloud Engineer -- Practice Scenarios

GCP Study Notes

Study Notes April 2026

28 realistic mini case studies covering all 5 exam sections. Use these to test your understanding, not just memorization. Each scenario is designed to mirror the style and difficulty of real exam questions.

Section Map

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Scenario 1: Startup With Unpredictable Traffic

Situation: A two-person startup is launching a new REST API for their mobile app. They expect traffic to be extremely variable – potentially zero requests at 3 AM, but spikes of thousands of requests during marketing campaigns. The team has no experience with Kubernetes and wants to minimize both cost and operational burden.

Requirements: - Scale to zero when there is no traffic - Handle sudden traffic spikes without manual intervention - Deploy containerized Go application with 4 HTTP endpoints - Keep costs as low as possible when idle

Question: Which compute service should the startup use?

► Click to reveal answer

Scenario 2: Legacy Windows Application Migration

Situation: A manufacturing company is migrating a legacy Windows-based ERP system to Google Cloud. The application requires Windows Server 2022, specific .NET Framework 4.8 libraries, and connects to a local SQL Server database via named pipes. The application cannot be refactored or containerized due to vendor restrictions.

Requirements: - Windows Server 2022 operating system - Full control over OS configuration and installed libraries - Persistent local storage for temporary processing files - Application runs 24/7 serving internal users

Question: Which compute service should they use, and what machine configuration is most appropriate?

► Click to reveal answer

Scenario 3: Batch Video Rendering

Situation: A media production company runs video rendering jobs that take 2-6 hours each. They submit 50-100 jobs per day during the work week but nearly zero on weekends. Each job can be checkpointed every 15 minutes, and if interrupted, can resume from the last checkpoint. The company is highly cost-sensitive.

Requirements: - Minimize compute cost for long-running batch jobs - Jobs must tolerate interruptions and resume from checkpoints - Cluster of VMs should scale based on the job queue - Jobs need high CPU count (32+ vCPUs per job)

Question: What compute configuration would minimize costs while meeting these requirements?

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Scenario 4: Microservices With Complex Orchestration

Situation: An e-commerce company has 15 microservices that need to communicate with each other. The team uses Kubernetes extensively in their on-premises data center and wants to replicate this architecture in the cloud. They need rolling updates, automatic pod healing, service discovery, and the ability to run GPU-based recommendation workloads alongside standard web services.

Requirements: - Kubernetes-native orchestration with 15+ services - GPU node pool for ML inference workloads - Rolling updates with zero downtime - Team already proficient with kubectl and Helm

Question: Which compute platform should they use, and should they choose Standard or Autopilot mode?

► Click to reveal answer

Scenario 5: Image Thumbnail Generator

Situation: A photo sharing platform needs to generate thumbnails whenever a user uploads an image to Cloud Storage. Each thumbnail generation takes about 5 seconds. The platform sees 10,000-50,000 uploads per day, spread unevenly throughout the day. The thumbnail function is a simple Python script that reads the uploaded image, resizes it, and writes the result back to a different bucket.

Requirements: - Triggered automatically when a new image is uploaded to Cloud Storage - Simple, single-purpose processing (resize image) - No infrastructure management - Cost-effective for variable workload

Question: Which compute service and trigger mechanism should be used?

► Click to reveal answer

Scenario 6: Global Financial Trading Platform

Situation: A financial services company is building a trading platform that must serve users across North America, Europe, and Asia. The system handles millions of transactions per day, requires strong consistency for all reads (a trader must never see stale data), and must maintain 99.999% availability. The data is relational with complex join queries across orders, accounts, and positions tables.

Requirements: - Global distribution across three continents - Strong consistency for all reads and writes - Relational data model with complex SQL joins - 99.999% availability SLA - Millions of transactions per day

Question: Which database service should they choose?

► Click to reveal answer

Scenario 7: IoT Sensor Data Pipeline

Situation: An agriculture technology company deploys 50,000 soil sensors across thousands of farms. Each sensor sends temperature, moisture, and pH readings every 30 seconds. The data must be ingested at low latency, stored for real-time dashboards (sub-second queries), and also available for historical batch analysis. The data volume is approximately 2 TB per day.

Requirements: - Ingest millions of data points per minute - Sub-millisecond read latency for real-time dashboards - Time-series data optimized for range scans - Scale linearly as sensor count grows - Historical data available for batch analytics

Question: Which primary database should store the real-time sensor data, and how should historical analytics be handled?

► Click to reveal answer

Scenario 8: E-Commerce Product Catalog

Situation: A mid-sized e-commerce company needs to store product information (name, description, price, categories, variants, reviews) for 500,000 products. The web and mobile applications need to display product pages with sub-50ms response times. The data is semi-structured – different product categories have different attributes (electronics have specifications, clothing has sizes/colors). The company expects to grow to 2 million products within 2 years.

Requirements: - Semi-structured data with flexible schemas per product category - Sub-50ms read latency for product page rendering - Real-time updates when prices or inventory change - Mobile app needs offline capability and real-time sync - Data volume under 500 GB currently

Question: Which database service is the best fit?

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Scenario 9: Long-Term Compliance Archive

Situation: A healthcare company must retain patient record backups for 7 years due to regulatory requirements. The data is generated monthly (approximately 500 GB per month) and is almost never accessed – only in the rare event of a legal or compliance audit (estimated once per year or less). When accessed, a retrieval delay of a few seconds is acceptable.

Requirements: - Minimize storage costs over 7 years - Data must be retained and cannot be deleted before the retention period - Retrieval within seconds when needed (not hours) - Approximately 500 GB added per month

Question: Which storage class and configuration should be used?

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Scenario 10: Multi-Region Web Application Load Balancing

Situation: A SaaS company serves customers globally from the United States, Europe, and Asia. They have Compute Engine instances running their web application in us-central1, europe-west1, and asia-east1. They want users to be automatically routed to the nearest healthy region, with SSL termination and DDoS protection. Some URL paths (`/api/*`) should go to the API backend, while `/static/*` should be served from Cloud Storage.

Requirements: - Global traffic distribution to the nearest healthy region - URL-path-based routing to different backends - SSL/TLS termination at the load balancer - DDoS protection - Single anycast IP address

Question: Which load balancer should they use, and what additional services should be integrated?

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Scenario 11: Secure Hybrid Cloud Connectivity

Situation: A retail company has an on-premises data center with critical inventory and POS systems. They are extending to Google Cloud for new workloads. The connection must be encrypted, provide 99.99% SLA, and support dynamic routing as they add new subnets in both environments. Expected bandwidth usage is about 500 Mbps during peak hours. Budget is moderate – they cannot justify the cost and lead time of a dedicated physical connection.

Requirements: - Encrypted connection between on-premises and GCP - 99.99% availability SLA - Dynamic routing (BGP) to automatically learn new routes - Bandwidth needs up to 500 Mbps - Cost-effective setup within days, not weeks

Question: Which connectivity option should they choose?

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Scenario 12: Cross-Project Resource Communication

Situation: A company has two GCP projects: project-frontend (web application) and project-backend (API services and databases). VMs in project-frontend need to communicate with VMs in project-backend over internal IP addresses. The networking team wants to keep traffic on Google's private network, avoid traversing the public internet, and ensure IP ranges do not overlap. They do not need transitive connectivity to any third network.

Requirements: - Communication between VMs in two different projects using internal IPs - Traffic must stay on Google's private network - Simple setup without organizational complexity - No need for transitive connectivity

Question: Which networking solution should they implement?

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Scenario 13: Zero-Downtime Web Server Fleet Update

Situation: A company runs a fleet of 20 identical web servers using a Managed Instance Group behind an External Application Load Balancer. They have a new version of their application baked into a new custom image. They want to update all instances to the new image with zero downtime. They also want to test the new version with a small percentage of instances first.

Requirements: - Update all 20 instances to the new image - Zero downtime during the update - Test with 20% of instances before full rollout - Automatic rollback capability if health checks fail

Question: How should they perform this update?

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Scenario 14: Canary Deployment for API Service

Situation: A fintech company runs a payment processing API on Cloud Run. They have a new version that changes the fraud detection algorithm. They want to send only 5% of production traffic to the new version initially, monitor error rates and latency for 24 hours, and then gradually increase to 100% if metrics are healthy. They also want a dedicated URL to test the new revision without affecting production traffic.

Requirements: - Deploy new version without immediately receiving production traffic - Test the new revision via a dedicated URL - Gradually shift traffic: 5% then 25% then 100% - Instant rollback if issues are detected

Question: How should they deploy and manage traffic for the new version?

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Scenario 15: GKE Cluster With Mixed Workload Types

Situation: A data science company runs a GKE cluster with two types of workloads: (1) web API services that need to be always available and scale with traffic, and (2) batch ML training jobs that are cost-sensitive and can tolerate interruptions. They want to optimize costs by using cheaper nodes for the batch jobs while keeping the web services on reliable nodes.

Requirements: - Web services on reliable, always-available nodes - Batch ML jobs on cost-optimized nodes that may be reclaimed - Prevent batch jobs from being scheduled on web service nodes - Automatic scaling of nodes based on pending pods

Question: How should they configure the GKE cluster?

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Scenario 16: Developer Access Control for Multiple Teams

Situation: An organization has three development teams: Frontend, Backend, and Data. Each team should only manage resources relevant to their work. Frontend developers need to deploy to Cloud Run. Backend developers need to manage Compute Engine VMs and Cloud SQL instances. Data engineers need to run BigQuery queries and manage Cloud Storage buckets. No team should have access to modify IAM policies or billing.

Requirements: - Frontend: deploy and manage Cloud Run services - Backend: manage VMs and Cloud SQL - Data: query BigQuery and manage Cloud Storage - No team should modify IAM or billing - Use the principle of least privilege

Question: Which IAM roles should be assigned to each team, and how should access be organized?

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Scenario 17: Secure VM Without External IP

Situation: A security-conscious bank runs application VMs that must not have external IP addresses (enforced by organization policy `constraints/compute.vmExternalIpAccess`). Developers still need SSH access to these VMs for troubleshooting. The VMs also need to download security patches from the internet and access Cloud Storage APIs.

Requirements: - VMs must have no external IP addresses - Developers must be able to SSH into VMs - VMs must access the internet for package updates - VMs must access Google Cloud APIs (Cloud Storage, etc.) - All access must be auditable

Question: What combination of services enables SSH access, internet access, and API access for VMs without external IPs?

► Click to reveal answer

Scenario 18: Service Account for Automated Pipeline

Situation: A CI/CD pipeline running on a Compute Engine VM needs to build Docker images, push them to Artifact Registry, and deploy them to Cloud Run. The security team insists on following least privilege and prohibits the use of service account keys. The VM currently uses the default Compute Engine service account, which has `roles/editor`.

Requirements: - Push container images to Artifact Registry - Deploy services to Cloud Run - No service account keys - Least privilege (remove excessive permissions from default SA) - Auditable

Question: How should the service account be configured?

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Scenario 19: Temporary Contractor Access

Situation: A company hires a contractor for a 3-month project ending March 31, 2026. The contractor needs to view and create Compute Engine instances in the `dev-project` only, and must be able to SSH into the VMs they create. They should not be able to delete instances, modify networking, or access any production projects. The company wants access to automatically expire.

Requirements: - Create and view VMs in dev-project only (no deletion) - SSH access to VMs - Access expires automatically on March 31, 2026 - No access to production projects or networking

Question: How should this access be configured?

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Scenario 20: Proactive CPU Monitoring and Alerting

Situation: A DevOps team manages 50 Compute Engine VMs running a distributed application. They have experienced outages when VMs silently max out CPU without anyone noticing. They want to be alerted when any VM's CPU utilization exceeds 85% for more than 5 minutes. Alerts should go to both a Slack channel and the on-call email address.

Requirements: - Alert on CPU > 85% sustained for 5+ minutes on any VM - Notify via Slack and email simultaneously - Include runbook instructions in the alert notification - Track the number of high-CPU incidents over time

Question: How should they configure this monitoring setup?

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Scenario 21: Automated Database Backup Strategy

Situation: A company runs a production Cloud SQL PostgreSQL instance that stores critical customer data. The compliance team requires: daily automated backups with 30-day retention, the ability to restore to any point in the last 7 days, and a tested disaster recovery procedure. The operations team also wants to receive alerts if backup storage exceeds a certain size.

Requirements: - Daily automated backups retained for 30 days - Point-in-time recovery within a 7-day window - Automated backup schedule without manual intervention - Alert if backup storage grows unexpectedly

Question: How should the backup strategy be configured?

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Scenario 22: Centralized Log Management With Compliance Requirements

Situation: A financial company has 10 GCP projects. The security team requires that all error-level logs from all projects are exported to BigQuery for analysis, all logs are retained for at least 2 years, and debug-level logs from load balancers are excluded from storage to reduce costs. They also need to identify who deleted any resource across the organization.

Requirements: - Export ERROR+ logs from all projects to BigQuery - 2-year retention for all other logs - Exclude load balancer debug logs to save costs - Track all resource deletions across the organization

Question: How should the logging infrastructure be configured?

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Scenario 23: Cost Optimization for Development Environments

Situation: A company has 5 development teams, each with their own GCP project. Monthly cloud spending across dev projects has grown to \$15,000. The CFO wants to reduce this. Upon investigation, you find: VMs are running 24/7 even though developers work only during business hours (8 AM - 6 PM weekdays), several VMs are oversized (using n2-standard-8 but only consuming 15% CPU), and many Cloud Storage buckets contain old build artifacts that have not been accessed in over 6 months.

Requirements: - Reduce VM costs for off-hours and weekends - Right-size oversized VMs - Automatically manage stale storage data - Set up budget alerts to prevent future overspending

Question: What cost optimization measures should be implemented?

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Scenario 24: Budget Alert With Automatic Spending Control

Situation: A startup received \$10,000 in Google Cloud credits. They want to be notified at 50%, 80%, and 100% of a \$8,000 monthly budget. Critically, they want to automatically disable billing on their project if spending exceeds \$9,000 to protect against runaway costs from a misconfigured service.

Requirements: - Budget alerts at 50%, 80%, and 100% of \$8,000 - Email notifications to the finance team - Automatic billing disablement at \$9,000 - Must work without manual intervention

Question: How should this be configured?

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Scenario 25: Multi-Environment Infrastructure With Terraform

Situation: A company needs to deploy identical infrastructure (VPC, subnets, firewall rules, Cloud SQL instance, GKE cluster) across three environments: dev, staging, and production. Each environment should be in its own project. The team wants version-controlled infrastructure, code review for changes, and the ability to preview changes before applying them. Five engineers collaborate on the infrastructure code.

Requirements: - Identical infrastructure definition for all three environments - Environment-specific configuration (project IDs, machine sizes, node counts) - Shared state accessible by 5 engineers with locking - Preview changes before applying - Version control and code review

Question: How should the Infrastructure as Code setup be organized?

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Scenario 26: Secure Containerized Application End-to-End

Situation: A healthcare startup is deploying a patient portal application on GKE. The cluster must have no public node IPs. Pods need to access a Cloud SQL PostgreSQL database over a private IP and read patient documents from a Cloud Storage bucket. The application images are stored in Artifact Registry. The security team requires that no service account keys are used anywhere and that the default service account must not be used.

Requirements: - Private GKE cluster (no public node IPs) - Access Cloud SQL via private IP (no public endpoint) - Access Cloud Storage from pods without SA keys - Pull images from Artifact Registry - No default service account usage

Question: What architecture and IAM configuration should be used?

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Scenario 27: Migrating and Modernizing a Monolithic Application

Situation: An insurance company runs a monolithic Java application on-premises that handles policy management, claims processing, and customer notifications. They want to migrate to GCP in phases. Phase 1 (immediate): lift-and-shift the monolith to GCP to decommission the data center. Phase 2 (6 months later): extract the notification service into a serverless component triggered by Pub/Sub. They need to plan the architecture for both phases now.

Requirements: - Phase 1: Run the existing Java monolith as-is on GCP - Phase 1: Minimal changes to the application - Phase 2: Notification service extracted as an event-driven component - Phase 2: Notification service should scale independently and cost nothing when idle - Monitoring and logging from day one

Question: What GCP services should be used for each phase?

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Scenario 28: Organization-Wide Security and Governance

Situation: A large enterprise is setting up Google Cloud for the first time. They have 200 developers across 15 teams, with dev, staging, and production environments. The security team has the following mandates: no VM should have an external IP in production, all data must stay in the EU, billing must be tracked per team, and all API actions must be auditable. They also need to prevent accidental project deletion by developers.

Requirements: - Organization-wide restriction: no external IPs on production VMs - All resources restricted to EU regions - Billing tracked per team - Full audit trail of all API actions - Prevent developers from deleting projects

Question: How should the resource hierarchy, organization policies, billing, and auditing be configured?

► Click to reveal answer